

Maxim Silin

Mathematician and Programmer

Warwickshire, UK, Right to work in UK. (Able to relocate)

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Mathematician and Programmer.

Area of expertise involves differential equations, geometry and physics.

Website development, python scripting and deployment experience.

PROFESSIONAL EXPERIENCE

09/2022 - 07/2026 - Mathematical Tripos degree BA + MMath- Uni of Cambridge.

BA 1st Class Honours achieved + Mmath Merit achieved.

Skills:

Lab work (1st year)

Computational Projects in Matlab and Python

Reports Write-up via LaTeX mostly.

Research (Mathematical)

Achieved 1st Class Honours in 3rd year exams. (Transcript available)

Achieved Merit in Masters MMath.

07/2024 – 09/2024, Software Internship – Toku, Singapore, SG.

Description: worked in Singapore as part of a Summer Internship in Programming.

Programmed a large number of rules in Node.js, to be fired as alerts with different severity levels. Worked with regex expressions and considered various approaches and benefits/drawbacks. Implemented a parser to aid in translating the rules, while ensuring correctness after translation.

Technologies used: Amazon Cloudwatch, Lambda, EC2, Node.js, and others.

AWS experience, exposure to Terraform.

2021 – 2024, Web-site developer, MyArtventures, UK

3-tier web application development for the artists' portfolio.

Functionality implemented: Gallery of pictures, Picture page, Order form, Admin portal - editing and uploading content (photo and text), Contact form linked to email notification system.

Technologies used: React.js, JS, HTML, SASS (CSS), Redux, Node.js with Express.js, rate-limiters and csrf protection, nodemailer (from npm) as a notification service for admin, jwt cookies for authentication, MongoDB Atlas (Cloud) to store text meta-data, postman for API testing. API included all the usual CRUD operations, authentication, protected endpoints, image upload and delete, rate limiters for all endpoints, and email service/contact form, error reporting service.

Cloud Deployment & Maintenance:

Hosted this on a Digital Ocean droplet, using tools such as:

Ubuntu, NGINX for reverse proxy port forwarding, subdomains, rsa keys, ssh, process management, Connection to external MongoDB.

09/2021 - 06/2022, 3-sphere CS A-level Project (<https://github.com/Maxicl14/A-spherical-world>), UK**Simulation of a 4D sphere**

Instructions for set up: code1/INTSTRUCTIONS_FOR_USE.

Description: Implementing mathematical calculations in a C++ like language (WebGL) to make a game where a rocket is traversing a 4-dimensional sphere on the sphere's surface (3D warped), seeing planets, and bumping into them (collision detection & prevention).

Technologies used: Spherical ray casting, Vertex Shaders (in WebGL), JavaScript/HTML, JSON files (to store object maps), Python (to generate models of planets).

See the github repo

2018-2022

- **Noughts and Crosses game in Javascript & HTML, with 2 player or play against program.**
- **Wonky Chess game in Javascript & HTML.**
- **Various Projects in Javascript and WebGL.**
- **Aiding a startup in gamification of maths teaching project.**
- **Experimented with Contact centre (AWS connect) flow with customer data fetching and agent data saving.**

07/2017 - 09/2018, Python AES encryption research project, UK

Description: Used Python to program AES128 encryption algorithm. Used to encrypt a picture of my friend's face with AES128.

Technologies used: Python, AES.

2023-2025 Maths Tutoring, GCSE

Description: one-to-one tutoring of GCSE mathematics,

Private tutoring to prepare children for GCSE and expand their skillset and understanding. Weekly lessons and learning. Experience working with a child with Asperger's Syndrome (SEN).

EDUCATION

King Edward VI Stratford 6th Form

A levels: A* Maths, A* Physics, A* CS, A* Further Maths.

- Further Maths included mechanics and statistics courses.

Alcester Grammar School

- 10 GCSEs all nines including all sciences, English, Maths, Art, History, Computer Science, French.

- DofE Bronze and Silver – map reading, navigation.